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Lactate and Monocarboxylate Transporters (MCTs): A Review of Cellular Aspects

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ABSTRACT

Crisp AH, Verlengia R, Rocha GL, da Mota GR, Pellegrinotti IL, Lopes CR. Lactate and Monocarboxylate Transporters (MCTs): A Review of Cellular Aspects. **JEPonline** 2015;18(3):1-13. Lactate is a metabolite of the glycolytic pathway. Its concentration increases in the bloodstream in response to high-intensity exercises. Historically, high lactate production by muscle cells during intense physical exercise is associated with muscle fatigue and reduced physical performance. However, after the confirmation of the "cell-to-cell" and "intracellular" lactate shuttle hypotheses, the association of interference has been challenged by many researchers. Currently, lactate is considered an energy substrate reoxidized to pyruvate. Through the action of monocarboxylate transporters (MCTs), lactate contributes to the maintenance of intracellular pH. Therefore, the purpose of this review is to describe advances in the scientific views of the lactate molecule, its formation, function, and its transporters (MCTs) while also addressing the controversies.

Key Words: Lactate, Lactate Transporters, Physical Exercise

INTRODUCTION

The striated skeletal muscle is the largest lactate production site, with production triggered mainly in response to high intensity exercise. Since its discovery in 1780, a range of knowledge has been accumulated, showing the importance and complexity of the interaction of this metabolite with cellular energy demand.

Evidence indicates that lactate is one of the glycolytic pathway products originating via the action of lactate dehydrogenase (LDH) on pyruvate, which is facilitated across the plasma membrane of cells by the proton-linked monocarboxylate transporter (MCT) from the intracellular to the extracellular environment and vice versa by different isoforms. However, it is worth noting that some aspects of this process are still controversial. The need for further study and caution regarding the diffusion of knowledge on this subject is imperative. Among the various concerns are: (a) the association of lactate with muscle fatigue in response to exercising; and (b) the presence of MCT1 in mitochondria.

The purpose of this review is to summarize new information on the lactate molecule and its formation, function, and carriers with a comprehensive reflection on the established knowledge and present controversy. The survey was conducted using the databases Science Citation, Index, Scopus, Sport Discus, The SciELO and National Library of Medicine, with the combination of the following keywords: lactate; lactate shuttle; monocarboxylate transporters; MCT1; MCT4; and exercise biochemistry.

LITERATURE REVIEW

Historically, since the discovery of lactic acid by the chemist Carl Wilhelm Scheele in 1780, isolated from sour whey samples (49), high lactate production by muscle cells during intense physical exercise has been associated with muscle fatigue and reduced physical performance. However, due to the pKa value (constant acidity) of the lactic acid molecule being 3.86, physiologically lactic acid would be completely deprotonated. The loss of the proton causes the lactate molecule to become negatively charged, attracting positively charged ions to form a salt (sodium lactate) (48). Thus, use of the term "lactic acid" is inappropriate, since, even during high-intensity exercise intracellular and extracellular physiological pH levels are not low enough (3.86) due to the buffer system and the biochemical structures of the molecules of lactic acid ($\text{CH}_3\text{-CHOH-COOH}$) and lactate ($\text{CH}_3\text{-CHOH-COONa}$) are different.

In this regard, a review of the biochemical aspects of the production of lactate conducted by Robergs, Ghiasvand, and Parker (49) pointed out that the increase in the formation of H^+ during high-intensity exercise occurs by the high rate of ATP hydrolysis (mainly from the recruitment of type II fibers) and the H^+ would not be formed during the production of lactate from the LDH enzyme action. Barker, McCormick, and Robergs (3) consider lactate production essential for the maintenance of high intensity exercise. According to them, it is necessary to remove the accumulation of pyruvate and to regenerate cytosolic NAD^+ to continue the reaction of glyceraldehyde-3-phosphate dehydrogenase (GAPDH) in order to maintain the functioning of the glycolytic pathway. Moreover, the reduction of pyruvate to lactate helps to maintain intracellular pH since the LDH enzyme consumes a middle proton reducing pyruvate to lactate. The formation of lactate causes a decrease in intracellular acidosis. Therefore, H^+ and lactate would be produced by independent routes without a quantitative ratio (49).

Despite these relatively new discoveries, lactate production by skeletal muscle in response to intense exercise is still controversial. Several researchers, like Böning et al. (5) and Lindinger and colleagues (37), pointed to the relationship between the production of "lactic acid" and/or lactate with acidosis induced by high-intensity exercise. Moreover, Bangsbo and Juel (2) support the idea that lactate accumulation during intense exercise contributes to the development of fatigue. Yet, a series of discussions conducted by the editorial board of the Journal of Applied Physiology indicate the existence of contradictory views regarding the influence of lactate accumulation on muscle activity. For example, numerous researchers (32,33,35) share the idea that the accumulation of intramuscular lactate does not necessarily correspond to muscle fatigue and that other physiological mechanisms are more closely associated with fatigue. This debate, which has been reviewed by several leading researchers, confirms different biochemical points of view (15,41,47,50). Thus, there is still no consensus among exercise physiologists and other researchers regarding the possible acidosis associated with the accumulation of the lactate molecule.

Despite the muscle fatigue mechanism being a complex and multifactorial process, a decrease in pH causes decreased muscle contraction efficiency that results in fatigue (31). Thus, what can be highlighted is that intracellular acidification impairs the function of proteins and enzymes important in the process of muscle contraction due to: (a) the reduced operation of Ca^{2+} pumps and Ca^{2+} channels in the sarcoplasmic reticulum; (b) the alteration of the conformation of the enzyme ATPase; and (c) the inhibition of the activity of phosphofructokinase and glycogen phosphorylase enzymes. Ultimately, singly and collectively, each one reduces the contribution of the glycolytic system in cellular energy production.

Therefore, independent of the discussion, the lactate molecule *per se* and/or other mechanisms are responsible for increasing free intra and extracellular H^+ . Buffering mechanisms and the removal of H^+ are essential for the maintenance of high intensity muscle contraction for greater periods, thus resulting in better athletic performance. Currently, a major finding about the lactate produced by muscle cell functions is that it can be used as an energy source in different tissues (20,23). Hence, various investigations elucidate the location and function of MCTs in skeletal muscle and other tissues.

LACTATE/ H^+ TRANSPORTERS IN STRIATED SKELETAL MUSCLE

The striated skeletal muscle is known as the main driving force in the production of lactate. This metabolite, which is formed in the glycolytic anaerobic phase, has an increased concentration in the bloodstream, especially during high-intensity exercise (55). Initially, it was believed that lactate produced within the muscle cell was transferred via simple non-ionic diffusion. However, after verifying that the lactate and pyruvate transport in human erythrocytes was strongly inhibited by alpha-cyano-4-hydroxycinnamic acid (CHC), it was demonstrated that a transport mechanism specific for monocarboxylate could be identified (21).

Such findings are consistent with the fact that the cells are enveloped by the plasma membrane, and that most water-soluble molecules (except for lipid soluble molecules or small molecules such as oxygen and carbon dioxide) present in the cytoplasm cannot move freely across the membrane by simple diffusion due to the hydrophobic character of the lipid bilayer that composes it. Thus, for lactate to have this integrating function (between cells), a facilitated transport system using plasma membrane-bound proteins is required. These proteins are identified as MCTs or solute carrier family

16 (SLC16 - solute carrier family 16), where different isoforms are expressed in specific tissues (20, 23).

It was during the mid-90s with special techniques such as molecular cloning and sequencing that the first MTCs were identified (17,18,45,57). Currently, there are 14 known members of this family in which MCT1 and MCT4 isoforms (expressed in striated skeletal muscle) carry important metabolic compounds such as pyruvate, lactate, and ketone bodies in co-electroneutral transport of a proton (H^+) together with a monocarboxylate (in a ratio of 1:1) by a facilitated diffusion mechanism. The translocation to the plasma membrane and correct functional activity of MCT1 and MCT4 isoforms is necessary for co-expression of auxiliary glycoprotein known as CD147 (28,20).

According to Halestrap and Price (22), all SLC16 (i.e., solute carrier) family members share a characteristic protein sequence of 12 transmembrane domains having hydrophobic helical structure. The N-terminal end, C-terminal, and a large loop between the sixth and the seventh transmembrane domain are present in the intracellular environment and possess hydrophilic characteristics. The sequence conservation between the various members of this family of carriers is greater in the lower transmembrane domains and hydrophilic regions (Figure 1).

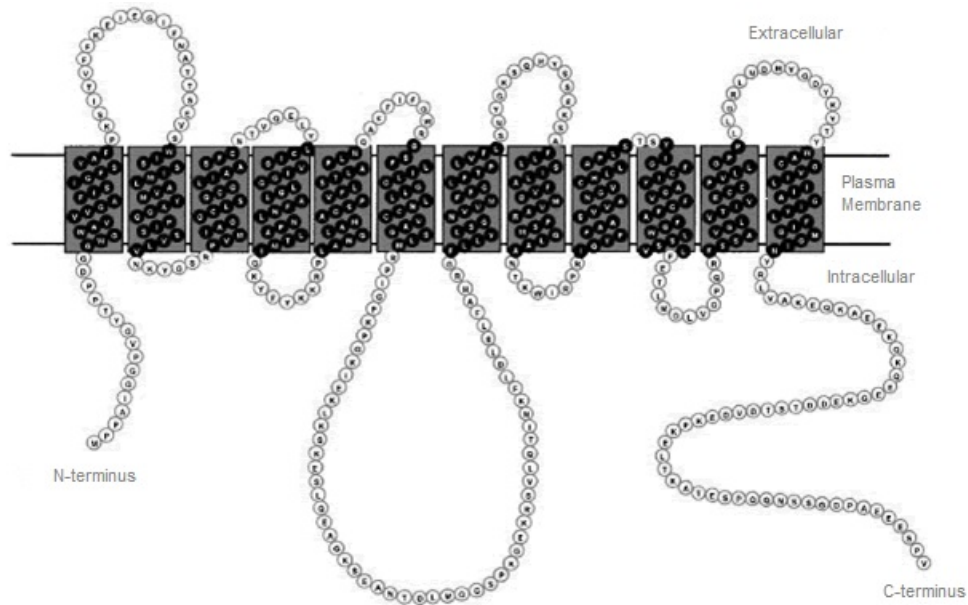


Figure 1. Topology Proposed for Members of the Family of Monocarboxylate Transporters (MCTs). The Sequence Shown is the Human MCT1 Protein. Modified from Halestrap and Price (22).

The muscle fibers (cells) are classified into different types according to biochemical, histochemical, and contractile characteristics (ATPase activity of the myosin heavy head). The presence of the MCT1 isoform in the plasma membranes of the fiber type I (oxidative trait) was identified in rats while the MCT4 isoform was found in a high percentage of type IIx fibers (glycolytic characteristic). Moreover, both proteins (MCT1 and MCT4) have been identified in type IIa fibers (glycolytic feature / oxidative) (34). In human skeletal muscle, Pilegaard et al. (43) reported that the MCT4 isoform was predominantly found in type II fibers with similar content MCT1 isoform between fiber types I and II, within a given cross section of a muscle biopsy of the soleus, triceps brachii, and vastus lateralis. Moreover, a positive correlation was found between total protein content of the MCT1 and the percentage of type I fibers (43).

Thus, the distribution of fiber types suggests that each MCT protein isoform may have distinct functions in the transport of monocarboxylics. The MCT1 operates predominantly with the influence of the lactate molecule from the circulation or extracellular medium from neighboring cells while MCT4 functions primarily for the efflux of intracellular lactate, especially in cells in which the production of lactate exceeds the oxidation of pyruvate. However, for Juel (30), the distinct roles proposed for the MCT1 isoforms (inflow) and MCT4 (outflow) are only based on different distributions of muscle fiber types, and that the transport of lactate/H⁺ is dependent on the concentration gradient. There is no evidence of a given isoform being specialized for transport in only one direction.

Interestingly, studies in rats have demonstrated that the expression of the MCT1 isoform in striated skeletal muscles is strongly correlated with the oxidative capacity of different types of fibers, enzymes citrate synthase, lactate dehydrogenase, cardiac isoform, with the rate of transport of lactate from the circulation, and inversely correlated with muscle fibers of glycolytic characteristics (39). In human vastus lateralis muscle biopsy, it was demonstrated that the MCT1 isoform protein content was reported as positively correlated to the rate of removal of blood lactate and inversely correlated with the fatigue index after various submaximal exercise sessions in cycle ergometer (53). Therefore, MCT1 would be related to the oxidative capacity and would allow better recovery between bouts of high intensity exercise.

The Km value (Michaelis-Menten constant, an indicator of the affinity for the substrate) of the MCT1 isoform is lower (3.5 to 8.3 mM) compared to MCT4 isoform (25 to 31 mM), indicating a higher affinity of MCT1 for lactate compared to MCT4. Thus, due to the high lactate transport capacity and low affinity, the MCT4 isoform is considered to be responsible only for the efflux of lactate/H⁺. As a result of the high Km value for the pyruvate molecule (>100 mM), the MCT4 isoform plays an important role for the muscle fibers with glycolytic characteristics, which is ensuring the conversion of pyruvate to lactate. This process is responsible for the re-oxidation of the NAD⁺ molecule and maintaining the glycolytic pathway (16).

CELL-TO-CELL AND INTRACELLULAR LACTATE SHUTTLE

Currently, lactate is recognized as a metabolite that can be used as an energy substrate by the skeletal muscle and other tissues (e.g., liver, heart, testicles, and ovaries). This occurs through the use of lactate shuttle, which has the function of transporting this molecule among different cells for oxidation. According to Brooks (6), the cell-to-cell lactate shuttle hypothesis was introduced at an international congress in Belgium, in August 1984 (First International Congress of Comparative Physiology and Biochemistry). The hypothesis states that lactate is not only one of the end products of the glycolytic system, it also has an important role in the distribution of energy among cells when reduced to pyruvate.

The concept of lactate shuttles relies heavily on the results of studies using isotopic labels to simultaneously determine the flow and lactate oxidation rates at rest and during exercise in rats and dogs (9,10,29) and, subsequently, in humans (38,52). According to Brooks (6), the results of these investigations indicate that oxidation was the main removal route of lactate from blood (~75%) during exercise and recovery in a stable state, with the balance (~25%) being removed by gluconeogenesis in the liver (i.e., the conversion of glucose to lactate). It is interestingly that the initial lactate shuttle hypothesis (7) occurred before the discovery of the first MCTs proteins and their distribution in different tissues (17,18,45,57). Moreover, the detection of the presence of the MCT1 mitochondrial isoform in the plasma membrane and of intra-mitochondrial lactate dehydrogenase enzyme (intra-

mitochondrial LDH) (12,13) led to a new interpretation of the monocarboxylate transporters where cytosolic lactate could be captured and re-oxidized in the mitochondria within the same cell (6).

Lactate dehydrogenase (LDH) is an enzyme composed of four subunits that control the conversion of lactate and pyruvate molecules. The subunits are composed of two types (muscle [M] and cardiac [H]) that are encoded by the genes *LDHA* (locus: 11p15.4) and *LDHB* (locus: 12.12.2-p14). Thus, combinations of different subunits allow for the formation of five isoforms: (a) LDH1 (composed of four H subunits); (b) LDH2 (composed of three H subunits and one M subunit); (c) LDH3 (composed of two H and two M subunits); (d) LDH4 (composed of one H and three M subunits); and (e) LDH5 (composed of four M subunits). Isoforms with predominance of the M subunit have higher expression in fibers with glycolytic feature, which reduce pyruvate to lactate. On the other hand, the most predominant H subunit isoforms have higher expression in fibers with oxidative characteristics, which oxidizes pyruvate to lactate (54).

In this regard, Brooks et al. (12) used Western blotting techniques and electron micrograph to assess mitochondria isolated from liver tissue, heart, and skeletal muscle of rats. They confirmed the presence of mitochondrial LDH. In their study, the oxidation of lactate and pyruvate molecules by muscle mitochondria was blocked by the MCT protein inhibitor (alpha-cyano-4-hydroxycinnamic acid), while the inhibitor of the LDH enzyme (oxamate) blocked the oxidation of the lactate molecule but not the pyruvate molecule. The results obtained by electrophoresis revealed the presence of the LDH1 isoform in the cytosol and in the mitochondrial membrane of cardiac muscle cells, while the LDH5 isoform was present in the cytosol and in the mitochondria membranes of liver and heart cells.

However, findings by Rasmussen and colleagues (46) and Sahlin et al. (51) strongly challenged the experiments and concepts that were proposed by Brooks et al. (12). Rasmussen et al. (46) isolated mitochondria from skeletal muscle of humans and rats to evaluate the possible role of the mitochondrial enzyme LDH and lactate as a substrate for the mitochondrial metabolism. The results of their study demonstrated the absence of activity of the LDH enzyme in mitochondria isolated from skeletal muscle and an inability to directly metabolize lactate. In addition, Sahlin et al. (51) isolated mitochondria from rat soleus muscle and reported an insignificant activity of the LDH enzyme in the mitochondrial fraction that represented only 0.7% of muscle activity. Their results showed that the mitochondria isolated from skeletal muscle easily oxidized the pyruvate molecule, but were unable to directly oxidize lactate. Thus, the data indicate that the oxidation of lactate can occur only after it is converted to pyruvate in the cytosol.

In a response letter to the editor of the *Journal of Physiology*, Brooks (8) criticized the methods of both studies (46,51), where they described extensive homogenization in the presence of proteolytic enzymes with protein agglomerates obtained by centrifugation and then treated with detergents. Consequently, the results included the loss of the LDH enzyme during the isolation of mitochondria. In 2004, the group led by Brooks (14) sought to confirm by Western blotting the location of the MCT1 protein in the long extensor muscles of the fingers, the soleus, and the heart of rats using three different cell fractionation techniques and three different antibodies. The study identified the presence of MCT1 in the mitochondrial membrane and the results were similar for the different methods of cell fractionation and antibodies used for analysis. The most likely reason for the differences in results between experiments can be attributed to procedures used to isolate the subcellular fractions.

In muscle cells, two mitochondrial populations are known for their subsarcolemmal (SS) and intermyofibrillar (IMF) locations and for having different characteristics. Benton et al. (4) examined the presence of the isoforms MCT1, MCT2, and MCT4 in purified samples of SS and IMF mitochondria in rat skeletal muscles. As expected, the presence of isoforms evaluated in the plasma membrane was

detected. However, while the MCT1 and the MCT4 isoforms were detected in the mitochondrial membrane of the SS portion, they were absent in the mitochondrial membrane of IMF. In contrast, MCT2 isoform (with high affinity for pyruvate) was present in both mitochondria. According to the authors, the selective presence of the MCT1 and the MCT4 isoforms in SS mitochondria indicates that lactate can be a monocarboxylic transported directly to the mitochondria, as initially proposed by Brooks. However, this was only true in mitochondria that were close to the plasma membrane.

In contrast, Hashimoto et al. (27), via histochemistry and immunohistochemistry of rat plantar muscles, revealed the presence of MCT1 in the sarcolemma and around the interior of the fibers with oxidative characteristics where the mitochondrial reticulum is present, while MCT4 was highly expressed in the sarcolemmal domain of glycolytic fibers. In addition, confocal laser scanning microscopy demonstrated that MCT1 proteins and cytochrome c oxidase (COX) were co-located in the IMF and SS cell areas, while MCT2 was weakly detected only in the sarcolemma of oxidative fibers. These results differ from those obtained by Benton et al. (4) who identified the MCT2 isoform only in IMF mitochondria.

With the use of new technologies, Hashimoto and colleagues (25) analyzed myocyte *in vitro* (L6 myocyte strain) derived from mice. They determined the location of mitochondrial LDH, MCT1, and CD147 by confocal laser scanning microscopy. In addition, Western blotting techniques of cell fractions and immunoprecipitation were used to confirm the results of a confocal scanning laser. The results demonstrated the presence of the proteins MCT1, CD147, and LDH in the mitochondrial reticulum. Western blotting analysis showed that COX NADH dehydrogenase, LDH, MCT1, and CD147 were abundant in mitochondrial fractions of L6 cells, which was confirmed by immunoblotting after immunoprecipitation interaction between COX proteins, MCT1, and CD147 in the mitochondria. The authors suggested the existence of a mitochondrial lactate oxidation complex associated with the complex IV in the electron transport chain. In summary, pyruvate is reduced to lactate by a process that generates NADH, which can be used in the electron transport chain where it is oxidized to pyruvate through the citric acid cycle (Krebs cycle). The mitochondrial lactate oxidation complex proposed by Hashimoto, Hussien, and Brooks (25) is shown in Figure 2.

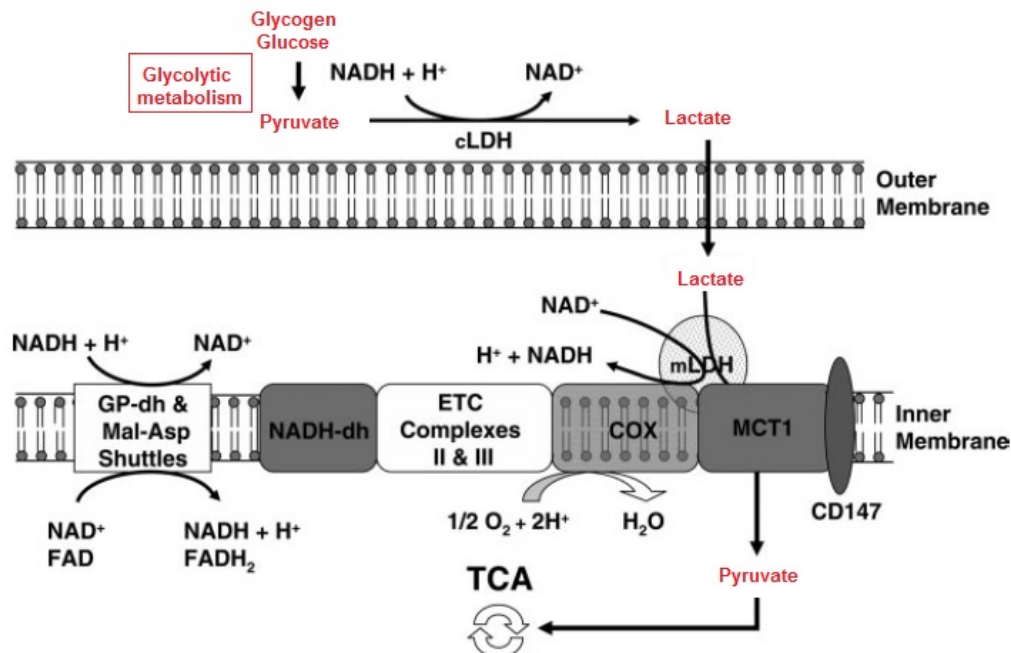


Figure 2. Schematic of the Mitochondrial Lactate Oxidation Complex Proposed in the Intracellular Lactate Shuttle Hypothesis. Modified from Hashimoto et al. (25).

In 2007, a group of investigators again found conflicting results regarding the mitochondrial oxidation of lactate (56). Western blotting analyses determined the presence of the MCT1 and MCT4 isoforms in SS mitochondria, but not in IMF mitochondria, while the MCT2 isoform was detected in both mitochondria types (IMF and SS), as previously found by Benton et al. (4). The activity of the LDH enzyme associated with the SS and IMF mitochondria was 200 to 240 times lower than the activity of the enzyme in the cytosol (muscle homogenates). In addition, lactate oxidation was significantly lower (31 to 186 fold) than pyruvate oxidation in SS and IMF mitochondria, as shown by radioactive tracer techniques. From these data, the authors concluded that the oxidation of lactate does not occur in the matrix of the SS and IMF mitochondria, due to the very limited amount of the LDH enzyme present in mitochondria.

The conflicting results are evident from the studies, as the hypothesis of direct mitochondrial oxidation of lactate was not confirmed by different laboratories (4,46,51,56), in addition to the studies conducted by the group led by Brooks (12-14,24,25,27). In the articles, the authors' were concerned regarding possible cross-contamination of other cellular components during the mixing of samples due to the fractionation and isolation of mitochondria from muscle cells. Thus, in addition to the analysis of MCTs, marker antibodies of the plasma membrane (GLUT4; Na^+/K^+ -ATPase), sarcoplasmic reticulum (Ca^{2+} -ATPase) and endosomal compartments (transferrin receptor) were evaluated, ensuring the purification of the mitochondrial samples. In a letter to the Journal of Physiology about the results obtained by Yoshida et al. (56), Brooks and Hashimoto (11) commented on several technical aspects of the study that may have led to not finding the expected results predicted by the intracellular lactate shuttle hypothesis. Among them, vigorous homogenization may have caused the loss of mitochondrial components and the isolation of mitochondrial fragments (IMF) involved treatments using proteases, which may have resulted in the loss of mitochondrial MCT1 and LDH proteins during this process.

Interestingly, in the same year, 2007, a group of Italian researchers (1) was the first, in addition to the studies led by Brooks, to identify the presence of the mitochondrial LDH enzyme by using primary cultures of granular cerebellar cells (CGCs) obtained from *Wistar* rats. Moreover, in 2008, an international group of researchers (42) used various contemporary techniques, including large scale mass spectrometry with the purpose of identifying the full spectrum of mitochondrial proteins (proteomics). The project, known as MitoCarta, identified a total of 1013 genes encoding human mitochondrial proteins including SLC16A1 (MCT1), LDHA, and LDHB. This content is available on the website: <http://www.broadinstitute.org/pubs/MitoCarta/human.mitocarta.html>.

In addition to mitochondria isolated from muscle cells, mitochondrial LHD was detected in astrocytes (36). Additionally, the mitochondrial lactate oxidation complex that was originally identified by Hashimoto, Hussien, and Brooks (25) in L6 myocytes mitochondria was also detected in rat neuronal cells (26). Therefore, an alternative model has been proposed for neuronal energy metabolism from the detection of the mitochondrial lactate oxidation complex. Currently, MCTs isoforms and lactate transport have also been studied in different tumors (44), since cancerous cells have glycolytic characteristics. It is suggested that lactate/ H^+ efflux is necessary to maintain the pH of cancer cells, to prevent the accumulation of this metabolite and to allow the maintenance of the glycolytic pathway (44). Thus, the selective inhibition of MCTs is a promising pharmacological strategy for cancer treatment (40).

FINAL CONSIDERATIONS

From its discovery to the present, there have been significant scientific advances in understanding the function and action of the lactate molecule. Currently, lactate is recognized as an energy substrate and, through the action of specific transporters, it contributes to the distribution of energy between tissues. At the cellular level, lactate is co-transported with the H⁺ proton by a facilitated diffusion mechanism from intracellular to extracellular space. Therefore, lactate is able to play an essential role in the maintenance of intracellular pH and promotes muscle contraction during high-intensity exercise. However, the association of lactate with muscle fatigue is not completely ruled out by the scientific community. The confirmation of the intracellular shuttle hypothesis supports the role of mitochondria as the organelle responsible for the reconversion and oxidation of lactate. Therefore, the measurement of blood lactate concentration can be considered as only an indirect measure of cellular metabolism, since part of the intracellular lactate is oxidized by the cell itself. Knowledge of the action of different MCTs isoforms in specific tissues highlights the complexity of the science of sport and exercise physiology regarding lactate.

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